

Math 8 Mid-Q4 Test — Answer Key

Mary, Star of the Sea School · Dr. Park · Answer Key (Teacher Use)

1. $V = 15\text{ V}$, $R = 3\ \Omega$. I ? **Answer:** 5 A
2. $I = 6\text{ A}$, $R = 4\ \Omega$. V ? **Answer:** 24 V
3. $V = 24\text{ V}$, $I = 6\text{ A}$. R ? **Answer:** 4 Ω
4. $P = VI$. $V = 5$, $I = 4$. P ? **Answer:** 20 W
5. $v = f\lambda$. $f = 6\text{ Hz}$, $\lambda = 4\text{ m}$. v ? **Answer:** 24 m/s
6. $v = 200\text{ m/s}$, $f = 20\text{ Hz}$. λ ? **Answer:** 10 m
7. If λ doubles (constant v), f ? **Answer:** halves
8. Higher frequency light $\rightarrow$ wavelength ? **Answer:** shorter
9. r doubles $\rightarrow T \times ?$ **Answer:** ≈ 2.83 ($\sqrt{8}$)
10. Earth $T=1\text{ yr}$. Planet $r=25\text{ AU}$. T ? **Answer:** 125 yrs
11. Planet $r = 16\text{ AU}$. T ? **Answer:** 64 yrs
12. Ohm's Law? **Answer:** $V = IR$
13. Wave equation? **Answer:** $v = f\lambda$
14. Higher $R \rightarrow I$ is ? **Answer:** lower
15. Kepler 3rd Law relates T to ? **Answer:** orbital radius
16. 100 W bulb for 20 hrs. Energy? **Answer:** 2000 Wh
17. 400 nm light in m? **Answer:** $4 \times 10^{-7}\text{ m}$
18. $v = 340\text{ m/s}$, $f = 340\text{ Hz}$. λ ? **Answer:** 1 m
19. Larger orbit = period is ? **Answer:** longer
20. Kepler's 3rd Law: $T^2 = ?$ **Answer:** kr^3